Macroeconomic Roles in Influencing the Correlation between Commodity Future with German Stock Index during German Economic Crisis

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List of Symbols

Symbol	Explanation
α	ARCH effect
β	GARCH effect
θ_1	The effect of recent shock on the current conditional correlation
θ_2	The effect of past correlation on the current conditional correlation
μ	The mean return of series
Ω	The base level of variance

1. Introduction

In recent years, financial institutions have expanded their investment portfolios towards commodity futures markets as an alternative investment in reducing their investment risk, especially during unpredicted shock. The increase inflow investment in commodity market, known as Financialization, has been recognized by the peak of average trading volume in the last decade (Bianchi et al., 2020), for instance: the annual trading volume on U.S. commodity futures markets in the year of 2021 had surged to new high at 40.6 trillion dollars (Kang et al., 2023). Outside U.S. market, China future market turnover increased by 6.28% year-over-year reaching 80 trillion US dollars in 2023 (Li, 2024).

The growing investment towards commodity future lead to a change in the interdependency between commodity futures and equities. This change can be associated with investor's decisions based on fundamental economic prices and macroeconomic information. It has been widely acknowledged that the stock market reflects the expectation of future economies. Likewise with commodity futures, given the commodity futures are forward looking investment, the investors always keep pay attention to macroeconomic information (Smales, 2017). Hence, the effectiveness of understanding the macroeconomic variables role on commodity futures and equities market are crucial to estimate the tendency of price movement. Besides investors, policy makers also take note on commodity market volatility given their potential to push inflationary pressures (Ahmed, 2023).

Existing literatures had investigated the economic application and co-movement between commodity futures and stock market. Liu and Guo (2022) employed shrinkage method to predict U.S. stock volatility using 19 commodity futures. Liang et al. (2020) used commodity futures in predicting U.S. stock market realized volatility (RV) by employing principle component analysis (PCA) and factor analysis (FA). The recent report by Wadud et al. (2023), they investigated the co-movement between 22 commodity futures and equity index using thick pen measure of association (TPMA) approach. In term of macroeconomic role in influencing commodity future market, Smales (2017) investigated the impact of U.S. and Chinese macroeconomic information on the volatility of commodity future market. Despite these interesting approaches, to the best of our knowledge, there is still limited understanding in term of macroeconomic variables role affecting this co-movement especially during German economic turmoil. Thus, this paper fills this gap. This paper also is the first one in employing DCC-GARCH approach to investigate the time varying correlation between commodity futures and German stock market index. Other study using DCC-GARCH method was adopted by Ding et al. (2021). They analyzed the dynamic correlation between commodity futures market and US stock market, represented by S&P 500 index.

This paper contributes to this literature by not only aiming exclusively on the dynamic correlation between commodity futures and German market, but also, we broaden the scope of explanation to which macroeconomics variables that potentially influence their dynamic correlation during German economic crisis. In this paper, we focus on four major crises that Germany has experienced for the last two decades, namely Post-2000 stagnation period (2000-2003), Global financial crisis (2008-2009), Euro area sovereign debt crisis (2009-2013), and COVID pandemic (2019-2020).

We are interested on German economy due to the fact that Germany heavily depending on commodities for industrial activity and food production, for example Russian natural gas. The supply chain disruptions of natural gas can have vulnerable impact on German economy by the rise in energy prices. Higher energy prices can lead to negative business expectation across German economy. Hence, investor sentiments toward slow German economy will affect the energy futures prices (Fasanya et al., 2022). It should be noted as well, as the third largest economy in the world behind United Stated and China and the largest economy in the Europe, Germany response to global economic downturn create a level uncertainty among investor and portfolio managers.

In the theory section, we explain the most recent literature describing relationship between macroeconomic variables and commodity futures volatility. Referencing recent publications, we identify which macroeconomic variables that influence commodity futures prices and stock market, the characteristic of commodity futures, and its relationship with spot prices. Not only that, we describe more details regarding the relationship between macroeconomic variables and stock market volatility during financial turmoil.

In the empirical analysis we investigate the time varying correlation between commodity futures and stock market analyze for the last 20 years. Our sample spans from January, 1st 2000 to December, 30th 2020. We include one German market index (DAX40) and three commodities futures (crude oil, natural gas, and wheat). Time varying multivariate DCC-GARCH model is employed to capture the dynamic correlation between market index and each commodities future during economic turmoil. This model allows us to observe the evolving correlation in nature between assets where it is impossible to capture by the static framework.

Our finding shows that there are persistence shock and correlation affect the conditional correlations between commodity futures and German stock index. The conditional correlation of each assets varies throughout the period of observation. During post-2000 stagnation period, the inverse relationship between DAX and natural gas is associated with the increase demand for natural gas and Investor sentiment towards weak economic growth. Strong demand on commodity due to lower interest rate kept upward pressure on commodity future prices during Global Financial Crisis. Stagnation period of German economy in the period of Euro area sovereign debt crisis are characterized by slow GDP growth rate and higher unemployment rate. The disturbance in supply chain during COVID pandemic scale back the company production, at the same time the reduction in export and import make investor sceptic towards commodity futures. Commodity future prices reflects a forward-looking expectation, making it responsive to change in fundamental and macroeconomic information.

The rest of this paper is organized as follows. The second section describing the theory of macroeconomic variables, commodity futures prices, and stock prices. The third section presents the methodology of DCC-GARCH model and describe the diagnostic test for standardized residuals. The fourth section describe the data and preliminary observation. The fifth section share the empirical result and elaborates the relevant discussion. The last section concludes this paper.

2. Theory on macroeconomic variable, commodity futures, and stock market

2.1. Macroeconomic variables and commodity futures

Investigating the factors influencing commodity futures volatility is important to better understand its pricing mechanism. Several macroeconomics variables, such as interest rate (Cornell & French, 1986; Gouel et al., 2024), money supply (Cornell & French, 1986), and exchange rate (Shang et al., 2016) had been shown to have impact on commodity futures prices.

As commodity futures market is characterized as a "forward looking", where its prices are attached to expectation on spot prices (Ye et al., 2021). It is critical to understand the source of commodity futures volatility for portfolio optimization and risk management. de Jong et al. (2022) had investigated how a commodity futures market affects spot price stability based on expectation of producers and speculators. They discovered that the expectations futures markets have a stabilizing effect on spot prices. Moreover, the variance of spot prices decline as interdependent between spot and futures markets becomes stronger.

Monetary policy impacts on commodity futures prices has been acknowledged thorough different mechanism. A higher interest rate tends to increase the opportunity cost of holding inventories, for instance oil inventories held in tanks. This leads to the reduction of demand and price for commodities. In addition to that, a higher interest rates increase the carrying cost of speculative positions, making high risk to bet on commodities. This will put downward pressure on futures price as well as spot prices (Frankel, 2007).

According to Chen et al. (2010), there are two mechanism of the dynamic relationship between exchange rate and commodity prices. The first mechanism explained that the speculative exchange rate can predict commodity prices based on net present value. The second mechanism (less robust), commodity prices can lead to change in exchange rate of the corresponding commodity currency, which is initiated by an increase in price of commodity. Higher price generates an upward pressure on demand for its currency which lead to an appreciation of exchange rate. Moreover, there was causal relationship between commodities prices (crude oil, gold, and copper) and exchange rate in both direction across multiple horizon in open economies with floating exchange rate countries, such as Canada, Australia, Norway, and Chile (Zhang et al., 2016).

2.2. Macroeconomic variables and stock market

There are substantial evidences that the behavior of stock market is dictated by macroeconomics factors, for instance money supply. Changes in monetary policy has direct impact on money supply and how investors made a decision on their investment. Hirota (2023) argues that the fluctuation of stock prices in the market is positively related to money supply, rather than influenced by change in the fundamentals, such as corporate earnings and economics conditions. The increase in amount of money available to investor increase the demand for the stocks and push stock price up. Using cointegration and vector error correction model, Bhuiyan and Chowdhury (2020) discovered that money supply has positive relationship with stock market, while the long term interest rate showed a negative relationship with stock market in the US. Using similar cointegration approach, US stock prices are positively influenced by industrial production and negatively by inflation and long interest rate (Humpe & Macmillan, 2009).

Extensive studies have been conducted to analyze the drive of stock market volatility. By constructing two-layer interconnected network between high-frequency stock price volatility and investor sentiment, Gong et al. (2022) found that stock price volatility of consumer goods is influenced by investor sentiment. Zhang et al. (2023) highlighted that geopolitical risk has a significant impact on stock market volatility. Using quantile-quantile dynamic approach, crudes price volatility has negative impact on the stock return in China, Japan, The United States, France, and Germany (Liu et al., 2023). These evidences suggest that the deep integration of global economic and the information transmission across financial market can influence stock market performance.

2.3. The connectedness between commodity future and stock market

There are the rapidly-growing investors interest in commodity futures, due to the benefit of the portfolio diversification in the short and long term (Wadud et al., 2023). By analyzing the commodity futures and stock market correlations, it allows hedge fund manager and investor to optimize their portfolio and manage their risk. Failing to consider their correlations, it can lead to an inaccurate hedging strategy and underperform portfolio. It has been shown that there is a bi-directional spillover between crude oil futures and India, Malaysia, and South Korea stock market. For gold futures, the bi-directional spillover occurs with China and Philippines stock market (Thuraisamy et al., 2013).

Recent study by Biswas et al. (2024), they examined the volatility connectedness networks of 18 global stock markets and 5 major commodities, they discovered that crude oil, platinum, and wheat became net transmitter and net receiver depending on the period of geopolitical tension between Russia and Ukraine. In addition to that, Creti et al. (2013) analyzed the links between price returns for 25 commodities and stocks over the period 10 years (2001 until 2011), they found out that commodity and stock market were highly linked during financial turmoil in the period of 2007-2008. Notably after financial crisis 2007-2008, according to Demiralay and Ulusoy (2014) finding, there were significant correlation between energy, agriculture, and industrial metal commodity indices with the S&P500 index.

3. Methodology

To analyze the time-varying conditional volatility during period of German economic crisis between market index and commodity futures, we employ Dynamic Conditional Correlation–Generalized Autoregressive Conditional Heteroskedasticity (DCC-GARCH) (Gabauer, 2020; Zhang et al., 2022). According to Engle (2002), this model has the flexibility of univariate GARCH model to describe how the correlations between different variables behave over time. We follow two step approach in DACC-GARCH model: firstly, we use GARCH model to estimate the conditional variance, and secondly, we use the results from the first step to estimate the conditional correlations that vary through time.

The multivariate DCC-GARCH model can be outlined as follows (Engle, 2002):

$$y_t = c_t + \varepsilon_t \qquad \varepsilon_t | N_{t-1} \sim N(0, H_t) \quad t = 1, \dots T \tag{1}$$

$$\varepsilon = H_t^{1/2} \upsilon_t \sim N(0, I) \tag{2}$$

$$H_t = D_t R_t D_t \tag{3}$$

$$D_t = diag\left(h_{11t}^{1/2}, \dots, h_{ZZt}^{1/2}\right) \tag{4}$$

y_t, c_t, ε_t and υ_t are $(Zx1)$ dimensional vectors standing for the analyzed time series, conditional mean, error term, and standardized error term, respectively. N_{t-1}indicates all information available from *1* to t – *1*. In addition to that, H_t, D_t, and R_t are (ZxZ) dimensional matrices that represent time-varying conditional variance–covariance, the time-varying conditional correlations, and the dynamic conditional correlations, respectively. Before we perform DCC-GARCH model, we employ Ljung-Box Q-statistic and ARCH-LM to observe autocorrelation and ARCH effect, respectively.

It is identified that the time series data contains missing value. The total number of missing values for German market index, future crude oil, future natural gas, and future wheat are 43 (out of 5369), 77 (out of 5185), 76 (out of 5180), and 88 (out of 5212) respectively. Before the raw data processes further, these missing data are simply omitted to avoid potential bias. Then, it is constructed to a continuous series of daily log returns for each asset.

$$rt = \log\left(p_{it}/p_{it-1}\right) \tag{6}$$

where rt is the daily log returns, and p_{it} indicates the closing prices at time t and t-1. (Mensi et al., 2024). We detect the variance changes of return by performing a modified iterated cumulative sums of squares (ICSS) algorithm (Inclán & and Tiao, 1994). Skewness and Kurtosis are performed based on D'Agostino (1970) and Anscombe and Glynn (1983) respectively. The normality distribution of data is verified using Jarque-Bera test. The stationarity of time series is examined using 'unit root test', namely Augmented Dickey–Fuller (ADF) and Kwiatkowsky, Phillips, Schmidt and Shin (KPSS) test. To check residual autocorrelation in multivariate DCC-GARCH model, the Hosking (1980) and McLeod and Li (1983) are performed.

4. Data and Preliminary Observations

This paper considers three commodity futures, namely crude oil (energy), natural gas (energy), and wheat (agro-commodity). DAX40 (Deutscher Aktienindex) is used to represent German market index. We use historical daily data in order to capture the transmission dynamic and the volatility commodity futures across period (Garcia-Jorcano & Sanchis-Marco, 2022). The data of equity market and commodity futures cover the period of January, 1st 2000 to December, 30th 2020. We divide the Germany national crisis into four period (Table 2). In addition to that, the commodity futures prices are converted from USD to EUR based on the daily exchange rate of each day observation to align with

DAX quote (Figure 1). The detailed data and exchange rate are downloaded from *Yahoo Finance*, that can be extracted from built-in updated R package "*Quantmod*".

Table 1. Market Index and Commodity Futures description

Ticker	Name	Sector	Description	Reference
DAX	DAX40	Market Index	German Market index of 40 most liquid German companies (quoted in EUR)	Deutsche Börse DBG (2021)
CO	Crude Oil	Energy	Crude Oil Future Contract (USD per Barrel)	Zhang and Hamori (2021)
NG	Natural Gas	Energy	Natural Gas Futures Contract (USD per Million Btu)	Geng et al. (2021)
WH	Wheat	Agriculture	Wheat Future Contract (USD per bushel)	Billah et al. (2023)

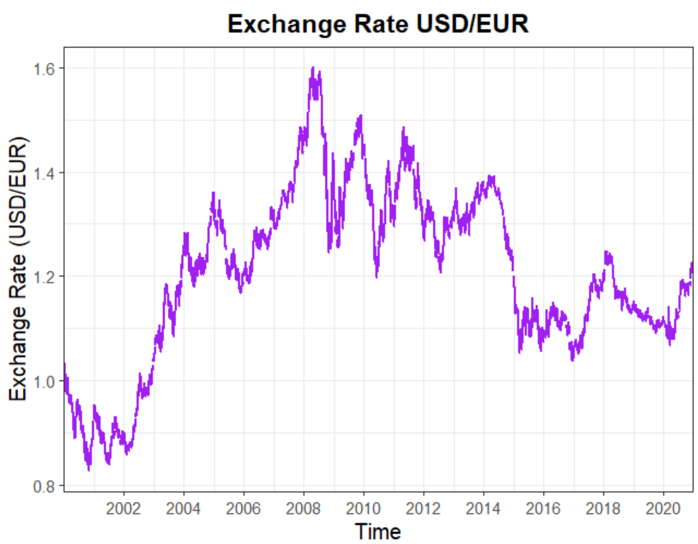

Figure 1. Daily exchange rate of USD/EUR from the period of 2000 until 2020

Table 2. German Economic Crisis

Crisis	Period	Reference
Post-2000 stagnation	2000 - 2003	Hein and Truger (2007)
Global financial crisis	2008 - 2009	Bessler et al. (2021)
Euro area sovereign debt crisis	2009 - 2013	Roman and Bilan (2012)
COVID pandemic	Late 2019 - 2020	Zhang and Hamori (2021)

As shown in Figure 2, the daily closing prices of DAX was falling as the dotcom burst in the year of 2000 until the year of 2003. Afterwards, it steadily increased with upward trend, but it dropped again in 2008 where it was known as the period of global financial crisis. Euro area sovereign debt crisis can be identified by the fall of DAX price in the beginning 2011. This dramatically decrease can also be observed during the COVID pandemic that broke out at the end of 2019. Overall, the trend of DAX daily closing prices shown an upward trend after the period of 2003. In contrast with CO, NG, and WH, the daily closing prices series exhibit unpredictable fluctuation and sudden price spikes for the last two decades. CO and WH show a distinction peak in the period of global financial crisis. The NG peak in the last period of 2005 was primarily driven by Hurricanes Katrina and Rita which disrupted NG production and distribution (Cruz & Krausmann, 2008).

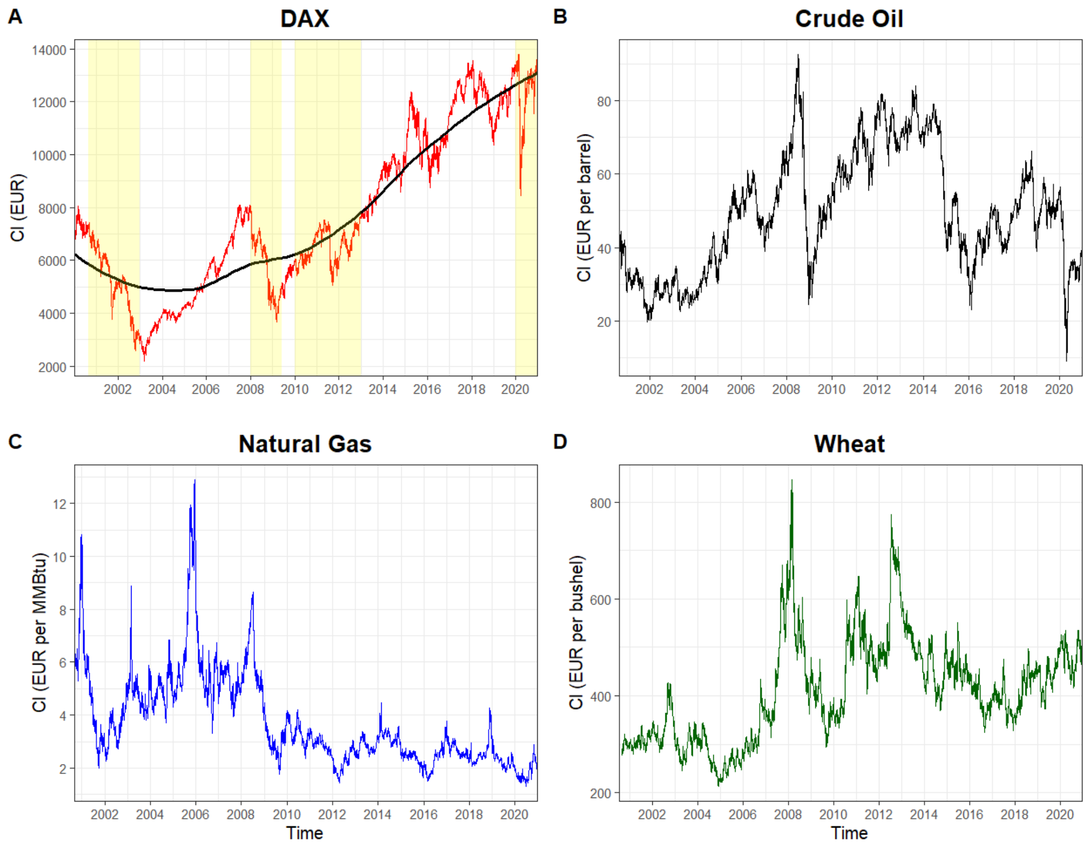

Figure 2. Closing prices (Cl) movement for market index and commodity futures. The shaded area in yellow represents the period of German Economic crisis. The black line indicates the trend line.

Figure 3 exhibits the daily return dynamics of DAX, CO, NG, and WH. Despite the closing price of DAX and CO are fluctuated during the sample period, the structural break is not detected among them. However, NG and WH experience significant volatility change at different period. A higher volatility entering the period of observation could be incorporated directly with the amount of natural gas in storage, resulting higher fluctuation in natural gas futures and spot prices (Chiou-Wei et al., 2014). The structural break in wheat futures contract is detected during global financial crisis. This circumstance lead to the non-convergence in wheat markets until the year of 2013 (Irwin, 2020).

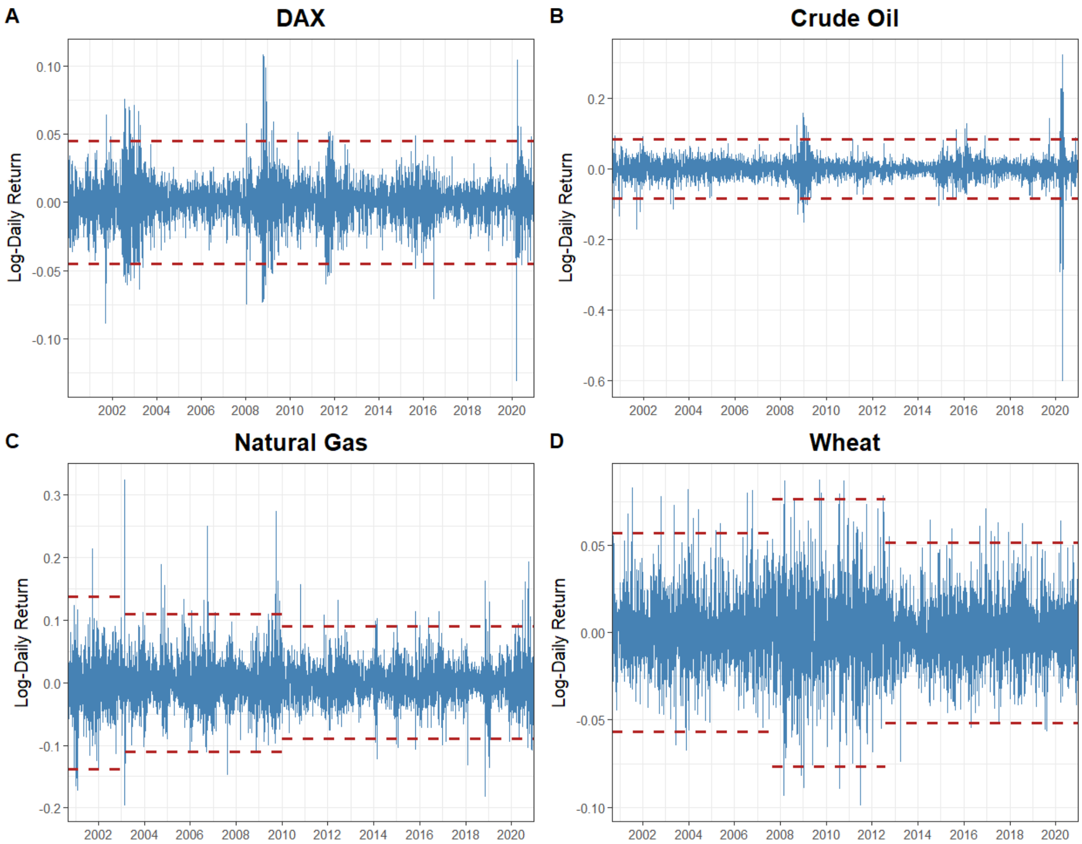

Figure 3. The log daily return of market index and commodity futures. The red bands represent the structural breaks estimated using modified ICSS algorithm at ±3 standard deviation.

Table 3 represents the descriptive statistics of the daily log return of DAX, CO, NG, and WH. The daily mean return is positive on DAX and WH, meanwhile CO and NG have negative daily mean return. The result of variance indicators shows that NG has the highest volatility while DAX has the lowest volatility. The negative skewness indicates that there is tendency to obtain negative extreme values of returns. The kurtosis value in all of assets exceeds the conventional standard of 3. This significance in excess kurtosis implies that all of the return series exhibit a leptokurtic distribution with fat tails compared to a normal distribution. This finding is supported by Jarque-Bera test, which implies that all series are not normally distributed. The unit root test ADF and KPSS shows consistent result. In ADF test, we reject null hypothesis which implies all of the return series are stationary. In conjunction with KPSS outcome, we fail to reject null hypothesis that confirm the stationarity of returns.

Table 3. Descriptive statistics

Statistics	DAX	CO	NG	WH
Mean	0.00011	-0.00005	-0.00012	0.00014
Variance	0.00022	0.00077	0.00118	0.00040
Minimum	-0.13054	-0.59900	-0.19583	-0.09879
Maximum	0.10685	0.32411	0.32411	0.08741
Skewness	-0.16402*	-2.00220*	0.51641*	0.18679*
Kurtosis	8.4191*	62.4090*	8.42060*	4.61830*
Jarque-Bera	6072.9*	730521*	6273.8*	568.4*
ADF	-16.72700*	-16.25900*	-16.37200*	-17.45000*
KPSS	0.07582	0.04455	0.01924	0.02382

Note: * indicates the statistical significance at the 1% significance level.

In Table 4, the Ljung-Box test at higher order shows the existence of autocorrelation in the return series and the squared returns. We reject null hypothesis of no serial correlation. In general, the serial correlation is common in financial time series data.

Table 4. Ljung-Box Q Autocorrelation test

Statistics	DAX	CO	NG	WH
LB-Q [30]	54.37*	81.94*	58.36*	29.13
LB-Q^2 [30]	4468.7*	1275.4*	559.03*	723.98*

Note: * indicates the statistical significance at the 1% significance level.

At different lag order, we reject the null hypothesis of no presence of ARCH effects. Table 5 shows the existence of volatility clustering in all assets. Volatility clustering is characterized by large (small) changes tend to be followed by large (small) changes. Therefore, the application of GARCH based approach is suitable for modelling stylized facts, for instance volatility clustering, persistence in commodity futures returns, and time varying correlation.

Table 5. ARCH-LM test

Statistics	DAX	CO	NG	WH
ARCH-LM test [10]	880.8*	757.8*	208.0*	221.2*
ARCH-LM test [20]	969.5*	813.9*	231.7*	253.9*
ARCH-LM test [30]	999.0*	1018.6*	255.5*	278.2*

Note: * indicates the statistical significance at the 1% significance level.

5. Empirical Result and Discussion

The results from our primary model, presented in Table 6, shows GARCH term is significant in all of assets. This indicates that the volatility of DAX, CO, NG, and WH in the previous period affects their current conditional variance. Meanwhile, ARCH term is significant on NG and WH. This suggests that previous period shock on return of NG and WH affects their current conditional variance. The shape of volatility growth exhibit fatter tails than a normal distribution. The sum of the ARCH and GARCH terms are close to unity, which implies that the volatility of all assets displays high persistence. Our findings are consistent with previous finding by Kang and Yoon (2020), they reported that the past shock and persistence of volatility influenced the Chinese commodity futures contract, namely: aluminum, copper, and natural rubber. The sort run reaction to a shock "θ_1" and the long run persistence of correlations "θ_2" are highly significant. This indicates that a shock and previous correlation affect the conditional correlations. The diagnostic test of Ljung-Box Q test for standardized residual reveals no evidence of autocorrelation in the univariate GARCH model. In addition to that, Hosking and Li-McLeod test supports the evidence of no misspecification of the multivariate DCC-GARCH model.

Table 6. Parameters estimation of DCC-GARCH

Parameters	DAX	CO	NG	WH	Joint
Panel A. estimates of univariate GARCH model					
μ	0.000735*	0.000303	-0.000390	-0.000256	
Ω	0.000002	0.000012	0.000018*	0.000004*	
α	0.092965	0.058337	0.074831*	0.038336*	
β	0.901667*	0.658306*	0.911681*	0.952059*	
shape	7.733715*	7.643124*	6.858420*	7.845745*	
θ_1					0.007617*
θ_2					0.987989*
Panel B. Diagnostic test for standardized residual					
LB-Q [30]	29.87	15.911	35.448	20.326	
p-value	(0.4721)	(0.9835)	(0.2267)	(0.9078)	
LB-Q^2 [30]	34.385	41.98	15.72	26.09	
p-value	(0.2657)	(0.0718)	(0.9850)	(0.6704)	
Hosking [30]					499.980
p-value					(0.2552)
Li–McLeod [30]					499.999
p-value					(0.2552)

Note: * indicates the statistical significance at the 1% significance level.

We are able to capture the co-movement between DAX against each commodity futures in the phase of Germany national turmoil (Figure 4). The conditional correlation exhibits distinguish pattern through observation and experiences phase of decreases and increases according to market conditions. We observe the correlation particularly during the period of Germany national turmoil which are Post-2000 stagnation period (2000-2003), Global financial crisis (2008-2009), Euro area sovereign debt crisis (2009-2013), and COVID pandemic (late 2019-2020).

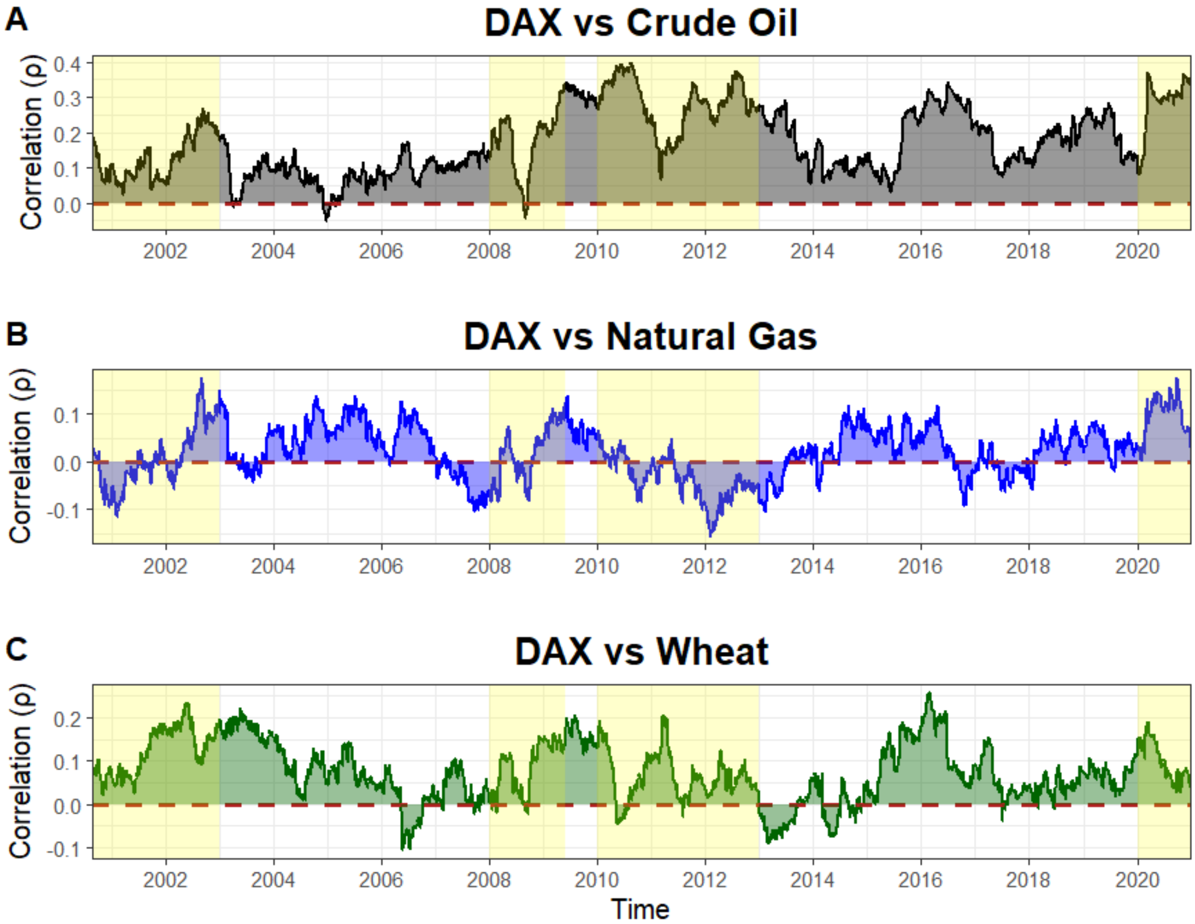

Figure 4. Time varying correlation between market index and each commodity futures. The shaded area in yellow represents the period of major crisis that German faced in the last 20 years.

In the post-2000 stagnation period, Germany experienced higher unemployment and lower average annual GDP growth around 0.7 percent (Hein & Truger, 2007). During this period, the pairwise conditional correlation of DAX and CO has similar pattern with DAX and WH (Figure 4). Both conditional correlations show positive correlation. In contrast with the conditional correlation between DAX and NG, the inverse conditional correlation is pronounced and reach the lowest point on the onset of 2001. This inverse relationship is triggered by the increase demand for natural gas due to higher demand in industrial consumption, such as the chemical industry, metal industry, building materials, pulp and paper and the food industry, resulting in subsequent increase in natural gas futures prices. Concurrently, Investor sentiment towards weak economic growth during post-2000 stagnation period lower the performance of DAX (Chen et al., 2025).

The global financial crisis emerging between 2008-2009 had stronger impact on German economy. It was reported that German GDP shrunk at 4.75% and unemployment rate increased at 7.5% (Schindler, 2013; Storm & Naastepad, 2015). The slowing down German economic was reflected by sharp decline of DAX performance (Figure 2). In term of the dynamic correlation (Figure 4), it exhibits convergence patterns where there is an abrupt drop towards negative correlation by the end of 2008 in all of assets. Afterwards, the magnitude correlations increase towards positive correlation. This result is in accordance with the finding of Öztek and Öcal (2017). They suggested that there was an upward trend correlation and high volatilities driven by financial crises between agricultural commodity sub-index, precious metal sub-index and stock market (S&P500). Moreover, Silvennoinen and Thorp (2013) highlighted that the correlation between wheat futures and S&P500 significantly increased up to 0.5 during financial crisis. Crude oil future and S&P500 correlation also increased during financial crisis, but it remained low and constant for natural gas. From macroeconomic standpoint, the negative correlation by the end 2008 can be associated lower interest rate. The lower interest rate put upward pressure on commodity futures prices by promoting strong demand on global commodity futures. A lower interest rate reduces the opportunity cost of holding inventory and lower the carrying cost of speculative positions (Frankel, 2007). Therefore, these circumstances promote strong demand on commodity futures. For example: during global financial crisis, European Central Bank cut the interest rate by 325 basis points (ECB, 2009). DAX lost 7% each day in 2008 (Herrera & Schipp, 2014), at the same time global oil demand increase at the peak of 87.8 million barrels per day in the fourth quarter of 2007 (ECB, 2010).

Euro area sovereign debt crisis in the late 2009 was triggered by the unsustainable Greek public debt, leading to loss of investor confidence and fear of contagion effect with other European countries (Gómez-Puig & Sosvilla-Rivero, 2016). During this crisis, each commodity futures correlation with DAX react in different way (Figure 4). In the beginning of 2010, there is slightly increase of correlation between DAX and CO, afterwards it drops significantly. For DAX and NG, their correlation shows a distinct downward trend until 2012. DAX and WH correlation drop to negative correlation in early crisis then it fluctuates between correlation 0 and 0.2. The pronounced macroeconomic conditions throughout this crisis is that Germany experienced slow pace of GDP growth rate (0.5%) and high unemployment (6.9-7.4%) (Bundesbank, 2012; DW, 2012). The stagnation German economic promote uncertainty in the market. Investor perceive this situation as slow economic growth and less company earning, as a result the performance of German market index decreases. In parallel, investor diversify their portfolio in commodity market as a safe-haven asset in order to reduce risk weight (Dimitriou et al., 2020). Hence, this condition creates an inverse relationship between commodity futures and German stock index.

COVID pandemic starting in the late 2019 caused a downturn in global economic. It was characterized by a stagnant real economy and severe drop in companies earning. According to German Federal Statistic Office, German economy experienced deep recession with the GDP shrinking by 5.7% and consumption expenditure dropping by 6%. Our finding shows positive correlation in all assets, even though the correlation between DAX and WH decrease in the beginning period of pandemic (Figure 4). Throughout COVID pandemic event, Germany encountered supply chain disruption resulting lack of demand and intermediate input deliveries in all sectors. Ultimately, many companies suffered from the reduction of turnover. Investor considered this condition as a risky event, so they tend to be risk aversion during economic downturn. As a result, many investors were scrambling for cash, known as "dash for cash". These compound events drive DAX price down substantially (Figure 2A) (Papavassiliou & Xia, 2025). In conjunction with commodity market, lower demand in commodity can be observed by the dropping in export by 9.9% and imports of goods and services by 8.6%, resulting drop in commodity prices (Eurosystem, 2021). The efforts by German government to mitigate the impact of pandemic by set up a large and stimulus package appear not to be effective in the service sector (Hinterlang et al., 2023). Reduction in both the DAX performance and demand on commodity futures are most reasonable in determining the positive correlation among them.

6. Conclusion

The main purpose of this paper is to prove the pair correlation between German stock index and commodity futures contact during Germany economic crisis and to offer an explanation which macroeconomics information potentially affecting their dynamic correlations. The important result emerges from this study,

- A shock and previous correlation affect the conditional correlations among all assets. The conditional correlation of DAX with each asset varies throughout the period of observation.
- This inverse relationship between DAX and natural gas is associated with the increase demand for natural gas due to higher demand in industrial consumption, resulting in subsequent increase in natural gas futures prices. Concurrently, Investor sentiment towards weak economic growth during post-2000 stagnation period drops the DAX return.
- Lower interest rate leads to the increase demand on commodity future, as a result it keeps upward pressure on commodity future prices. DAX performance drop significantly during Global Financial Crisis. These compound events promote the negative correlation between DAX and each commodity futures.
- The negative correlation during Euro are sovereign crisis can be affiliated with lower performance of DAX due slowing down in German GDP growth rate and higher unemployment rate. At the same time, diversification portfolio towards commodity futures push its prices higher.
- The positive correlation during COVID pandemic is associated with the disturbance in supply chain scaling back the company production, together with drop in export and import making investor sceptic towards commodity futures.
- Commodity future prices reflects a forward-looking expectation, making it responsive to change in fundamental and macroeconomic information.

The argument of macroeconomic roles on this paper are constructed exclusively based on the relevant literature. Nevertheless, our finding not only can be used as a foundation to describe their dynamic correlation, but also it can be used as references to narrow the scope of macroeconomic variables influencing their correlation. Further empirical analysis should be conducted in quantifying the magnitude of each macroeconomic variable affects the relationship between commodity futures and German market index.

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